

The role of TMEM126A in mitochondrial calcium homeostasis and retinal organoid models of optic atrophy

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Mitochondrial diseases are a major cause of inherited retinal degeneration, due to the high energy demand and calcium sensitivity of retinal neurons. TMEM126A, a mitochondrial inner-membrane protein associated with autosomal-recessive optic atrophy, is required for the assembly of Complex I and may influence mitochondrial dynamics and calcium signaling, but its precise role in retinal pathophysiology remains unclear. In this project, we aim to test the hypothesis that TMEM126A mutations alter mitochondrial calcium homeostasis, thereby contributing to mitochondrial dysfunction in retinal tissue.

As a preliminary model, we are currently using hTERT-RPE-1 cells to probe the functional impact of TMEM126A modulation. Mitochondrial calcium dynamics are being assessed with an aequorin-based luminescent reporter, alongside morphometric analysis of mitochondrial network architecture, measurements of respiratory chain activity and overall bioenergetics. Preliminary findings indicate that TMEM126A silencing leads to marked mitochondrial fragmentation, suggesting a role in mitochondrial fission dynamics; however, it is still unclear whether TMEM126A directly affects calcium handling or the overall respiratory efficiency, and further experiments are ongoing to clarify these aspects.

Building on these findings, we will transition to a physiologically relevant human model by generating iPSC-derived retinal organoids carrying patient-associated TMEM126A mutations. This platform will allow us to investigate the impact of TMEM126A deficiency on retinal development, mitochondrial morphology, calcium homeostasis, and neuronal survival in a human-retina-like context. In parallel, we aim to develop a proof-of-concept gene therapy approach based on lipid nanoparticles (LNPs) to deliver corrective genetic constructs to the organoid system, potentially restoring TMEM126A function and improving mitochondrial parameters. By integrating mechanistic studies in RPE-1 cells with advanced iPSC-derived organoid models and LNP-based gene delivery, our project seeks to define the role of TMEM126A in mitochondrial calcium homeostasis and to pave the way for novel therapeutic strategies for mitochondrial retinopathies.

Keywords: TMEM126A, mitochondrial calcium homeostasis, optic atrophy, iPSCs-derived retinal organoids, lipid nanoparticles